

Keras

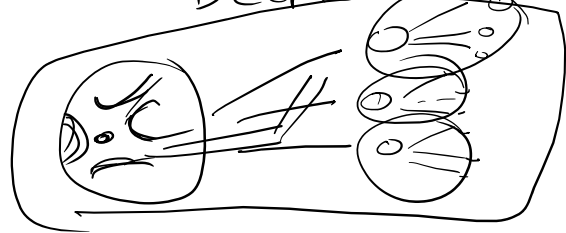
huge amount / unstructured data

image
video
audio

Complex problems

Deep learning

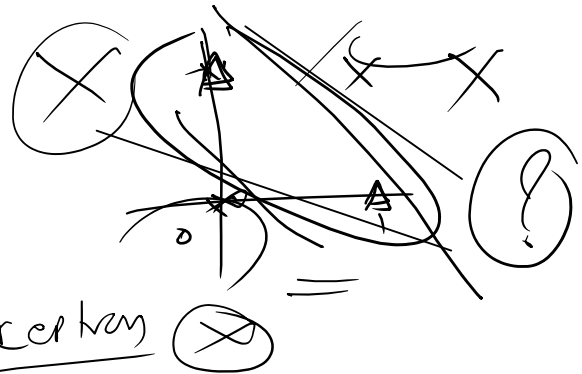
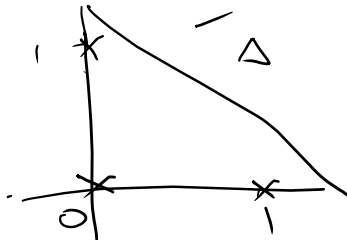
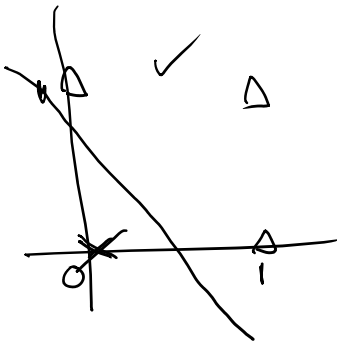
neuron, perceptron



A	B	$A \text{ OR } B$
0	0	0
0	1	1
1	0	1
1	1	1

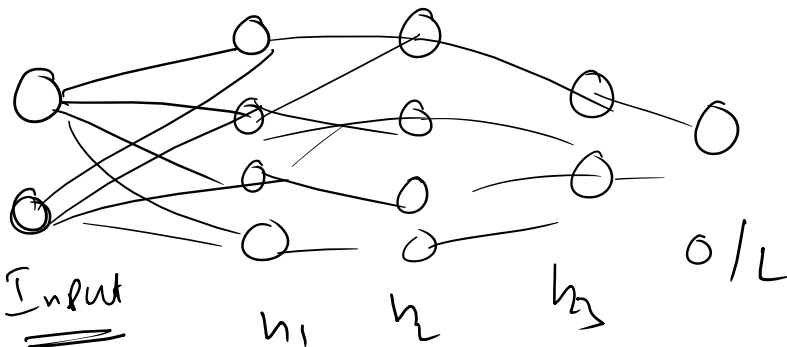
A	B	$A \text{ AND } B$
0	0	0
0	1	0
1	0	0
1	1	1

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

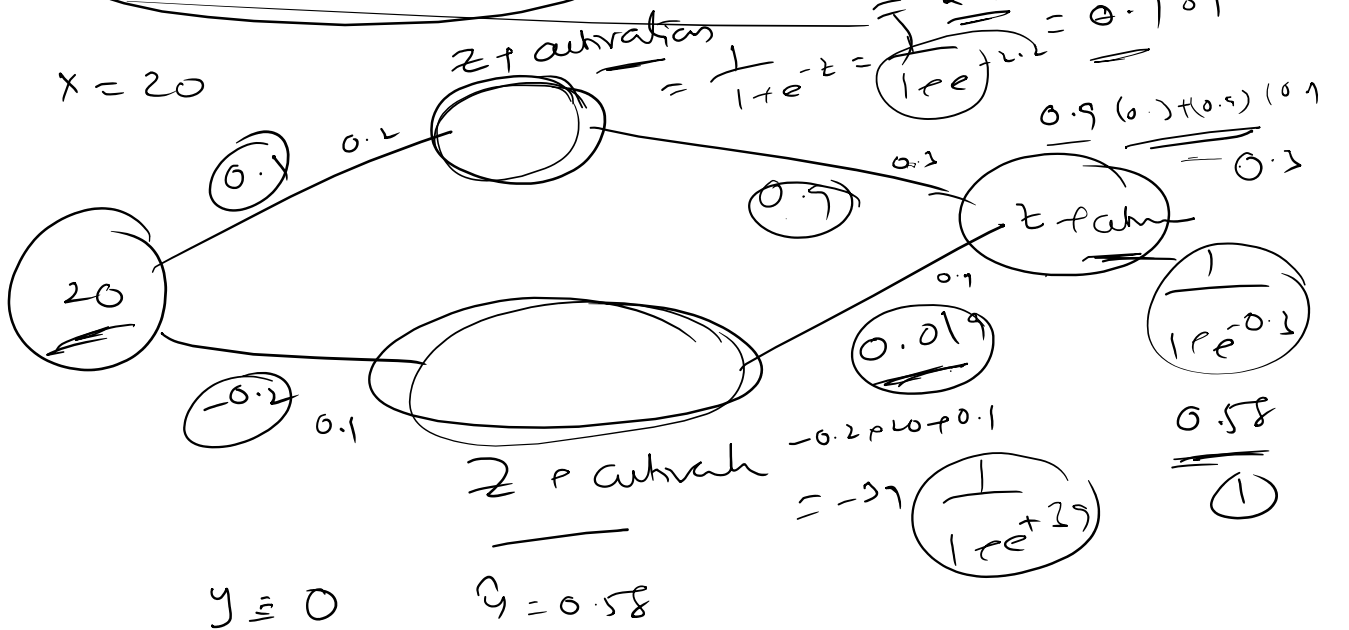


Perceptron

Neural network



$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



$$L = - [y \log y + (1-y) \log (1-y)]$$

$$= - [0 \log(0.58) + (1) \log(1-0.58)]$$

$$= 0.92$$

Random weight $\rightarrow$ prediction

low high

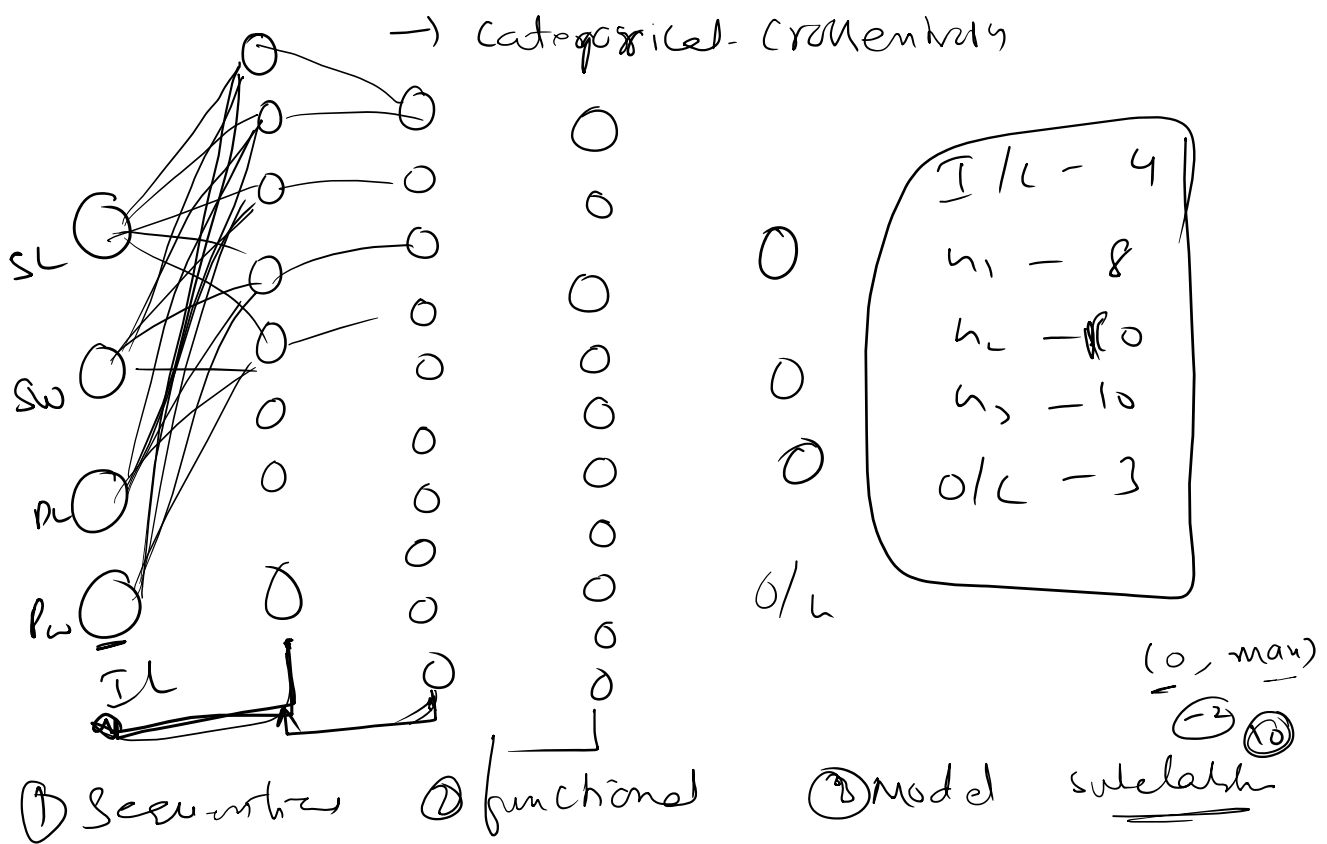
$\hookrightarrow$ update the weights

if the output is binary

$\rightarrow$ binary-cross entropy

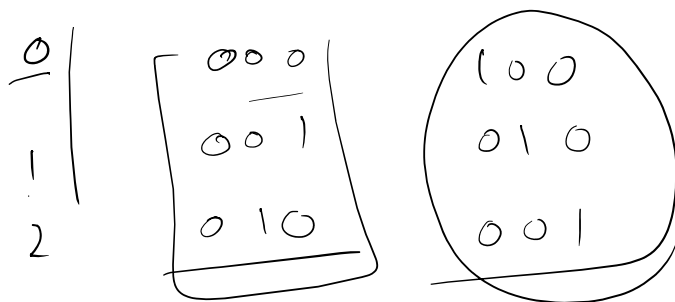
if the output is multiclass

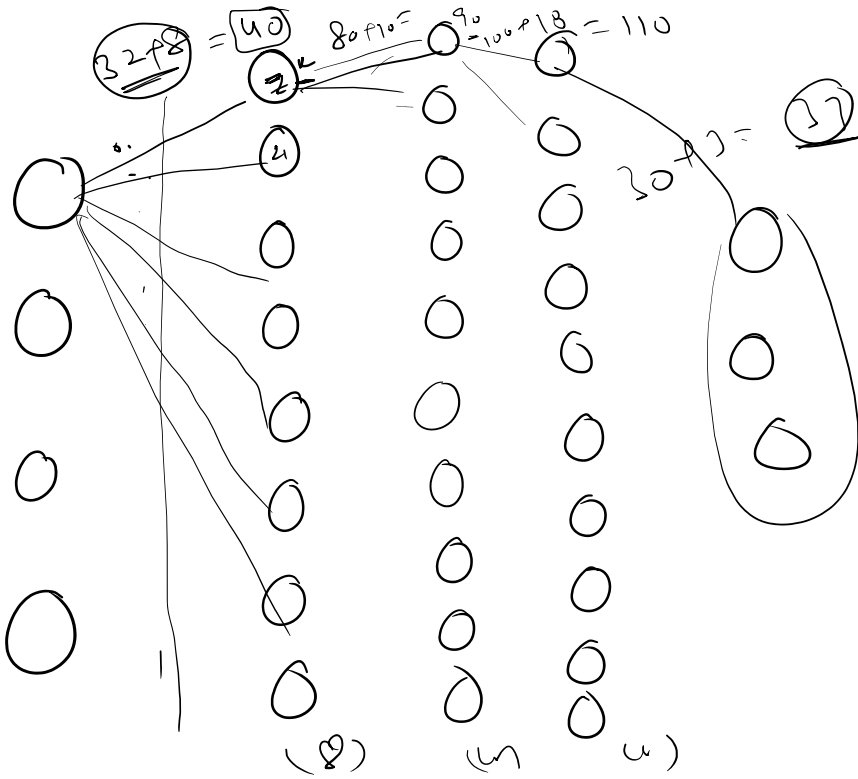
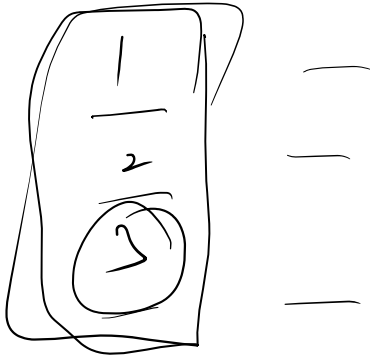
$\rightarrow$ categorical-cross entropy



```

model = Sequential()
model.add(Dense(8, input_dim=4, activation='relu'))
model.add(Dense(10, activation='relu'))
model.add(Dense(10, activation='relu'))
model.add(Dense(3, activation='softmax'))
  
```





$$\begin{array}{r}
 40 \\
 90 \\
 40 \\
 33 \\
 \hline
 2 \times 2
 \end{array}$$